

Election Analytics Platform for Portugal

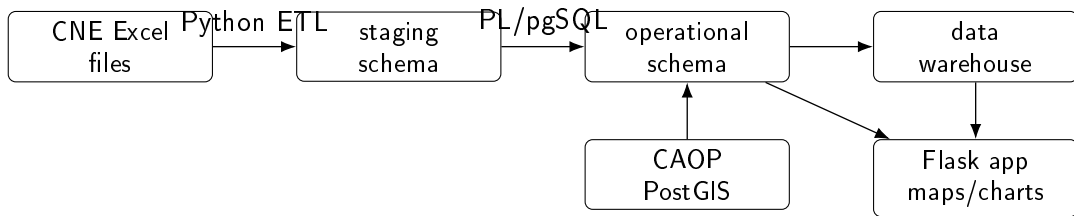
Advanced Topics in Databases Practical Assignment

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- Database-centred platform for analysing Portuguese election results.
- Datasets: official CNE Autarquicas 2013, 2017, 2021, and 2025 spreadsheet packages.
- Spatial layer: official DGT CAOP GeoPackages loaded into PostGIS.
- Thin Flask frontend using explicit SQL through psycopg2.

Architecture



- Core entities: elections, organs, territories, candidacies.
- Result entities: turnout, vote results, seat results, elected members.
- Territory hierarchy supports district/region, municipality, and parish drill-down.
- Candidacies model parties, coalitions, and citizen groups through a typed local candidacy table.
- Keys, constraints, uniqueness rules, and indexes support integrity and dashboard queries.

ETL Pipeline

- Rerunnable Python loader parses the CNE spreadsheets and populates staging.
- Loader is parameterized by source folder, election code, election name, and election date.
- PL/pgSQL procedure transforms staging rows into normalized operational tables.
- Warehouse refresh rebuilds dimensions and facts from the operational schema.
- CAOP loader reads GeoPackages, reprojects to EPSG:3763, and updates territory geometries.

Election	Vote rows
AL2013	12152
AL2017	12069
AL2021	12277
AL2025	12839

Warehouse and Analytical SQL

- Star schema: election, organ, territory, and candidacy dimensions.
- Central fact table stores votes, official percentages, mandates, and turnout measures.
- Analytical SQL includes:
 - D'Hondt mandate allocation;
 - three window-function queries;
 - ROLLUP and CUBE;
 - FILTER and JSON aggregation.
- Historical SQL compares turnout, vote-share trends, winner changes, and seat changes across 2013/2017/2021/2025.

- SQL functions: vote share, turnout rate, D'Hondt allocation.
- PL/pgSQL procedures: load operational schema from staging; refresh warehouse.
- Triggers enforce meaningful business rules:
 - no negative turnout/vote/mandate values;
 - blank plus null votes cannot exceed voters;
 - vote and seat rows must match the candidacy scope.

- PostGIS stores geometries as geometry(MultiPolygon, 3763).
- Loaded geometry coverage:
 - district/region: 20 / 20;
 - municipality: 308 / 308;
 - parish: 3241 / 3394 after the four-election load.
- Frontend includes election/organ selection, interactive Leaflet municipality map, result table, turnout summary, vote-share chart, seat chart, and historical comparison page.

- PostgreSQL validation checks staging, operational, warehouse, materialized views, and D'Hondt results.
- Four loaded elections: AL2013, AL2017, AL2021, and AL2025.
- Agueda municipal council D'Hondt matches official mandates:
 - PPD/PSD.MPT: 4;
 - PS: 2;
 - CDS-PP: 1.
- Flask endpoint tests:
 - homepage returned HTTP 200;
 - municipality results returned HTTP 200;
 - map endpoint returned 308 municipality features.

Limitations and Extensions

- Implemented extensions: historical comparisons, parish-level support, materialized views/precomputed aggregates, interactive map drill-down, and frontend comparison views.
- CAOP 2025 is used instead of CAOP 2021; it is official DGT data but not a perfect historical match.
- Parish geometries are partial because some codes changed across election cycles.
- Some elected-member rows do not match normalized parish candidacies exactly.
- Future work: multiple election types, election-type-specific frontend views, and improved parish reconciliation.

- 1 Show the repository structure and SQL/ETL files.
- 2 Run validation query for Agueda D'Hondt.
- 3 Open the historical comparison page for vote-share trends and winner changes.
- 4 Open the Flask map and select a municipality.
- 5 Show result table, turnout, vote-share chart, seat chart, and elected members.